In [2]:
```
import pandas as pd
dataset=pd.read_csv("Social_Network_Ads.csv")
dataset
```

Out[2]:

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0
...	...	...	...	...	...
395	15691863	Female	46	41000	1
396	15706071	Male	51	23000	1
397	15654296	Female	50	20000	1
398	15755018	Male	36	33000	0
399	15594041	Female	49	36000	1

400 rows × 5 columns

In [3]:
```
dataset=pd.get_dummies(dataset,drop_first=True)
dataset=dataset.drop(columns=["User ID"])
dataset
```

Out[3]:

	Age	EstimatedSalary	Purchased	Gender_Male
0	19	19000	0	True
1	35	20000	0	True
2	26	43000	0	False
3	27	57000	0	False
4	19	76000	0	True
...	...	...	...	...
395	46	41000	1	False
396	51	23000	1	True
397	50	20000	1	False
398	36	33000	0	True
399	49	36000	1	False

400 rows × 4 columns

In [4]:
```
x=dataset[["Age","EstimatedSalary","Gender_Male"]]
y=dataset[["Purchased"]]
print(y)
```

```
     Purchased
0            0
1            0
2            0
3            0
4            0
..         ...
395          1
396          1
397          1
398          0
399          1

[400 rows x 1 columns]
```

In [5]:
```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=1/8,random_stat
e=52)
dataset[["Purchased"]].value_counts()
dataset.shape
```

Out[5]:
```
(400, 4)
```

In [6]:
```
from sklearn.naive_bayes import MultinomialNB
clf = MultinomialNB()
clf.fit(x_train, y_train)
```

```
/usr/local/lib/python3.13/dist-packages/sklearn/utils/validation.py:1408:
DataConversionWarning: A column-vector y was passed when a 1d array was
expected. Please change the shape of y to (n_samples, ), for example using
ravel().
  y = column_or_1d(y, warn=True)
```

Out[6]:
```
MultinomialNB()
```

In a Jupyter environment, please rerun this cell to show the HTML representation or trust the notebook.
On GitHub, the HTML representation is unable to render, please try loading this page with nbviewer.org.

MultinomialNB
?Documentation for MultinomialNBiFitted

```
MultinomialNB()
```

In [7]:
```
y_pred=clf.predict(x_test)
print(y_pred)
from sklearn.metrics import accuracy_score
accuracy=accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy * 100, "%")
```

```
[0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 1 0 0 0 1 0 0 0 0 0 0 0 0 0
 0 0 1 0 0 1 1 0 1 0 0 0 0]
Accuracy: 70.0 %
```

In [8]:
```
from sklearn.metrics import confusion_matrix
r=confusion_matrix(y_pred,y_test)
print(r)
```

```
[[31 12]
 [ 3  4]]
```

In [9]:
```
from sklearn.metrics import classification_report
cr=classification_report(y_pred,y_test)
print(cr)
```

```
              precision    recall  f1-score   support

           0       0.91      0.72      0.81        43
           1       0.25      0.57      0.35         7

    accuracy                           0.70        50
   macro avg       0.58      0.65      0.58        50
weighted avg       0.82      0.70      0.74        50
```

In [10]:
```
from sklearn.naive_bayes import ComplementNB
clf = ComplementNB()
clf.fit(x_train,y_train)
```

```
/usr/local/lib/python3.13/dist-packages/sklearn/utils/validation.py:1408:
DataConversionWarning: A column-vector y was passed when a 1d array was
expected. Please change the shape of y to (n_samples, ), for example using
ravel().
  y = column_or_1d(y, warn=True)
```

Out[10]:
```
ComplementNB()
```

In a Jupyter environment, please rerun this cell to show the HTML representation or trust the notebook.
On GitHub, the HTML representation is unable to render, please try loading this page with nbviewer.org.

ComplementNB
?Documentation for ComplementNBiFitted

```
ComplementNB()
```

In [11]:
```
y_pred=clf.predict(x_test)
print(y_pred)
from sklearn.metrics import accuracy_score
accuracy=accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy * 100, "%")
```

```
[1 0 1 0 0 0 0 1 1 0 1 0 1 1 0 1 1 1 0 1 0 0 1 1 0 1 0 1 0 0 0 0 1 0 1 1 1
 0 0 1 1 1 1 1 0 1 1 0 0 1]
Accuracy: 46.0 %
```

In [12]:
```
from sklearn.metrics import confusion_matrix
r=confusion_matrix(y_pred,y_test)
print(r)
```

```
[[15  8]
 [19  8]]
```

In [13]:
```
from sklearn.metrics import classification_report
cr=classification_report(y_pred,y_test)
print(cr)
```

```
              precision    recall  f1-score   support

           0       0.44      0.65      0.53        23
           1       0.50      0.30      0.37        27

    accuracy                           0.46        50
   macro avg       0.47      0.47      0.45        50
weighted avg       0.47      0.46      0.44        50
```

In [14]:
```
from sklearn.naive_bayes import BernoulliNB
clf = BernoulliNB()
clf.fit(x_train,y_train)
```

```
/usr/local/lib/python3.13/dist-packages/sklearn/utils/validation.py:1408:
DataConversionWarning: A column-vector y was passed when a 1d array was
expected. Please change the shape of y to (n_samples, ), for example using
ravel().
  y = column_or_1d(y, warn=True)
```

Out[14]:
```
BernoulliNB()
```

In a Jupyter environment, please rerun this cell to show the HTML representation or trust the notebook.
On GitHub, the HTML representation is unable to render, please try loading this page with nbviewer.org.

BernoulliNB
?Documentation for BernoulliNBiFitted

```
BernoulliNB()
```

In [15]:
```
y_pred=clf.predict(x_test)
print(y_pred)
from sklearn.metrics import accuracy_score
accuracy=accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy * 100, "%")
```

```
[0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 0 0 0 0 0 0 0 0 0 0 0 0 0]
Accuracy: 68.0 %
```

In [16]:
```
from sklearn.metrics import confusion_matrix
r=confusion_matrix(y_pred,y_test)
print(r)
```

```
[[34 16]
 [ 0  0]]
```

In [17]:
```
from sklearn.metrics import classification_report
cr=classification_report(y_pred,y_test)
print(cr)
```

```
              precision    recall  f1-score   support

           0       1.00      0.68      0.81        50
           1       0.00      0.00      0.00         0

    accuracy                           0.68        50
   macro avg       0.50      0.34      0.40        50
weighted avg       1.00      0.68      0.81        50
```

```
/usr/local/lib/python3.13/dist-
packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning:
Recall is ill-defined and being set to 0.0 in labels with no true samples.
Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.13/dist-
packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning:
Recall is ill-defined and being set to 0.0 in labels with no true samples.
Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.13/dist-
packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning:
Recall is ill-defined and being set to 0.0 in labels with no true samples.
Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
```

In [ ]:
```
from sklearn.naive_bayes import CategoricalNB
clf=CategoricalNB(
    alpha=1.0, force_alpha=True, fit_prior=True,
    class_prior=None, min_categories=None
)
clf.fit(x_train,y_train)
```

In [ ]:
```
from sklearn.naive_bayes import GaussianNB
clf=GaussianNB()
clf.fit(x_train,y_train)
```

In [ ]:
```
y_pred=clf.predict(x_test)
print(y_pred)
from sklearn.metrics import accuracy_score
accuracy=accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy * 100, "%")
```

In [18]:
```
from sklearn.metrics import confusion_matrix
r=confusion_matrix(y_pred,y_test)
print(r)
```

```
[[34 16]
 [ 0  0]]
```

In [19]:
```
from sklearn.metrics import classification_report
cr=classification_report(y_pred,y_test)
print(cr)
```

```
              precision    recall  f1-score   support

           0       1.00      0.68      0.81        50
           1       0.00      0.00      0.00         0

    accuracy                           0.68        50
   macro avg       0.50      0.34      0.40        50
weighted avg       1.00      0.68      0.81        50
```

```
/usr/local/lib/python3.13/dist-
packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning:
Recall is ill-defined and being set to 0.0 in labels with no true samples.
Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.13/dist-
packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning:
Recall is ill-defined and being set to 0.0 in labels with no true samples.
Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.13/dist-
packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning:
Recall is ill-defined and being set to 0.0 in labels with no true samples.
Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
```